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B. Tech. III - Sem. (Main / Back) Exam., February - 2021

Aeronautical Engineering

3AN2-01 Advanced Engineering Mathematics – I

Common Aeronautical, Agriculture, Ceramic, Civil, ECE, EIC, ME, Mechatronics

Time: 3 Hours **Maximum Marks: 120**

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 10 × 2 marks = 20 marks. All ten questions are compulsory.*

Part – B: *Analytical/Problem Solving questions (up to 100 words) 5 ×8 marks = 40 marks. Candidates have to answer five questions out of seven.*

Part – C: *Descriptive/Analytical/Problem Solving questions 4 ×15 marks = 60 marks. Candidates have to answer four questions out of five.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

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PART - A

Q.1 Prove that $\mu^2 = 1 + \frac{\delta^2}{4}$

Q.2 State the fundamental theorem of finite differences and hence find the value of -

$$\Delta^6(ax - 1)(bx^2 - 1)(cx^3 - 1)$$

Q.3 Write the Newton's Forward Difference Interpolation formula.

Q.4 Prove that $\bar{F}[f(x - a); s] = e^{isa}\,\bar{F}[f(x); s]$

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Q.5 Write Euler's method to solve the ordinary differential equation.

Q.6 Find the Laplace transform of $F(t) = \frac{\cos h\, at}{\sqrt{t}}$.

Q.7 State and prove the second shifting theorem.

Q.8 Find $L^{-1}\left[\frac{1}{s(s^2+4)}\right]$

Q.9 Find the Fourier cosine transform of f (x), where $f(x) = \begin{cases} 1, & \text{for } 0 < x < a \\ 0, & \text{for } x > a \end{cases}$

Q.10 Find the z – transform of cos nθ and sin nθ.

PART – B

Q.1 Use Newton-Raphson method to find a real root of $f(x) = x^3 - 5x + 3 = 0$.

Q.2 Given the following pairs of values of x and y -

x:	5	6	9	11
y:	12	13	14	16

Interpolate the value of y at x = 10, using Lagrange's interpolation formula.

Q.3 Use Milne's Predictor – corrector method to find y (0.8). Given that $\frac{dy}{dx} = 1 + y^2$;

x:	0	0.2	0.4	0.6
y:	0	0.2027	0.4228	0.6841

Q.4 If $L\{f(t)\} = \bar{f}(s)$, then prove that $L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty \bar{f}(u)du$, provided that the integral exists.

Q.5 Find the Laplace transform of $\sin\sqrt{t}$. Hence show that $L\left\{\frac{\cos\sqrt{t}}{\sqrt{t}}\right\} = \sqrt{\frac{\pi}{s}}\, e^{-\frac{1}{4s}}$

Q.6 Find f (x) if its Fourier sine transform is $\frac{1}{s}e^{-as}$. Hence deduce $\bar{F}_s^{-1}\left(\frac{1}{s}\right)$.

Q.7 Find $z^{-1}\left[\frac{z}{(2-z)(2z-1)}\right]$

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PART – C

Q.1 (a) Compute the values of $\int_0^6 \frac{dx}{(1+x^2)}$ by -

(i) Trapezoidal rule

(ii) Simpson's $\frac{1}{3}$ rule

(iii) Simpson's $\frac{3}{8}$ rule

(b) The following table gives the values of a function at equal interval -

x:	1921	1931	1941	1951	1961
y:	19.96	39.65	58.81	77.21	94.61

Evaluate - (i) f (1936) (ii) f (1963) (iii) f (1951)

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Q.2 If $\frac{dy}{dx} = x + y^2$, use Runge-Kutta method to find an approximate value of y for x = 0.2. Given that y = 1 when x = 0. https://www.btubikaner.com

Q.3 (a) Prove that $L\left(\frac{\sin t}{t}\right) = \tan^{-1}\left(\frac{1}{s}\right)$ and hence find $L\left\{\frac{\sin at}{t}\right\}$. Does $L\left\{\frac{\cos at}{t}\right\}$ exist?

(b) Use Laplace transform theory to solve the following equation -

$(D^2 - 3D + 2)x = 1 - e^{2t}, x(0) = 1, x\ (0) = 0$

Q.4 Find the Fourier transform of $f(x) = \begin{cases} x^2, \text{when} |x| \le a \\ 0, \text{when } |x| > a \end{cases}$.

Hence evaluate - $\int_0^\infty \cos\left(\frac{as}{2}\right)\left\{\frac{(a^2s^2-2)\sin as + 2as\cos as}{s^3}\right\} ds$.

Q.5 (a) Use convolution theorem to evaluate -

$$L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$$

(b) Using z – transform solve the difference equation -

$$u_{n+2} - 3u_{n+1} - 4u_n = 3^n$$

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